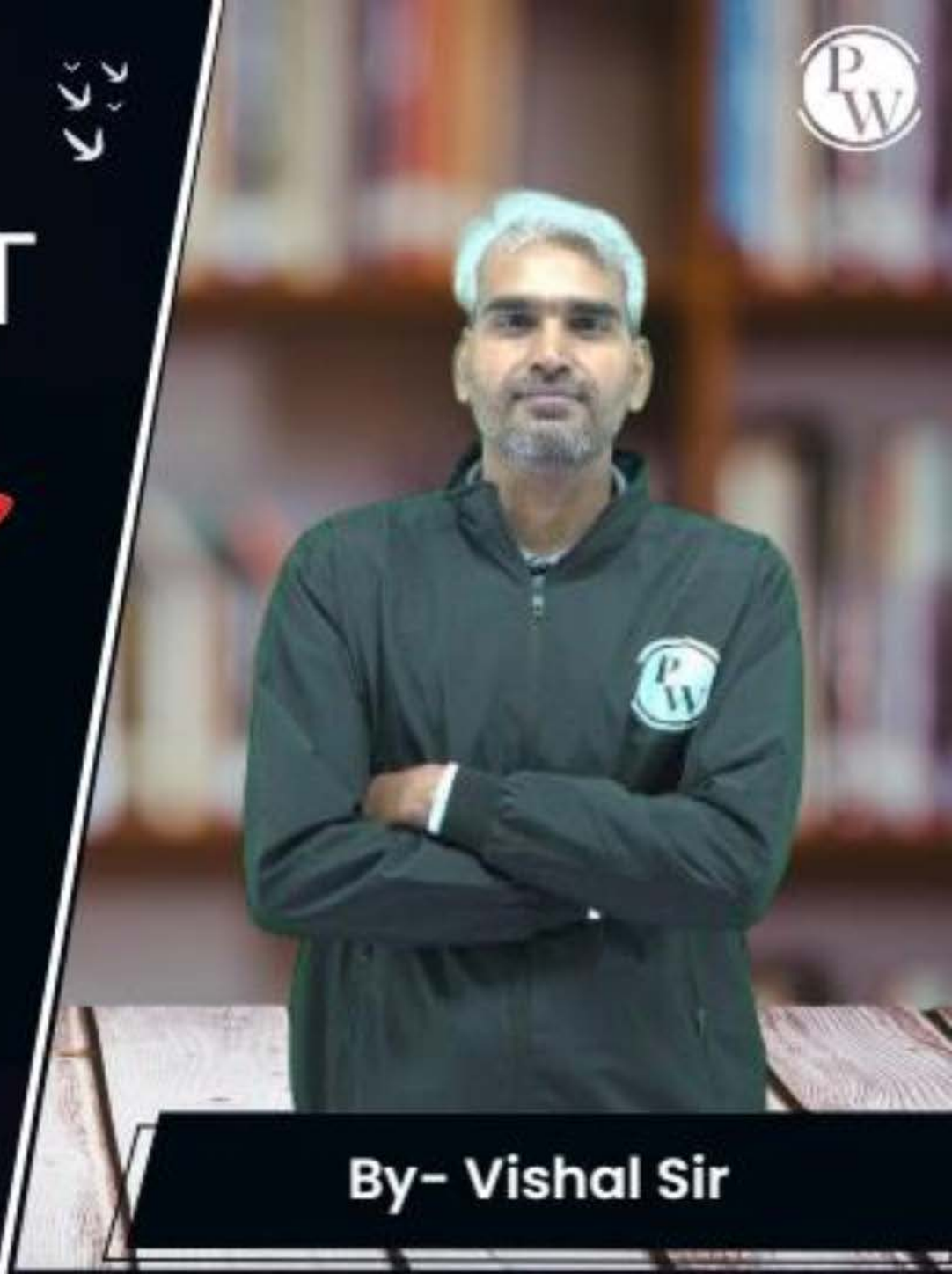
Computer Science & IT



Discrete Mathematics

Mathematical Logic

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Rules of inference



Slide

Topics to be Covered



- Topic — Practice questions
- Topic — Proof by contradiction
- Topic — Conditional proof
- Topic — Predicate logic

Slide

H.W.

Q:- Check whether the following argument is valid or invalid

- If today is Vishal's Bday, then today is 1st April
- Today is 1st April

∴ Today is Vishal's B'day

Let p = Today is Vishal's Bday

q = Today is 1st April

$p \rightarrow q$

q

$\therefore p$

is invalid

Q:- Check whether the following argument is valid or invalid

HW:

- If today is Vishal's Bday, then today is 1st April
- Today is not 1st April

∴ Today is not Vishal's B'day

p = Today is Vishal's Bday

~p = Today is not Vishal's Bday

q = Today is 1st April

~q = Today is not 1st April

$p \to q$

$\sim q$

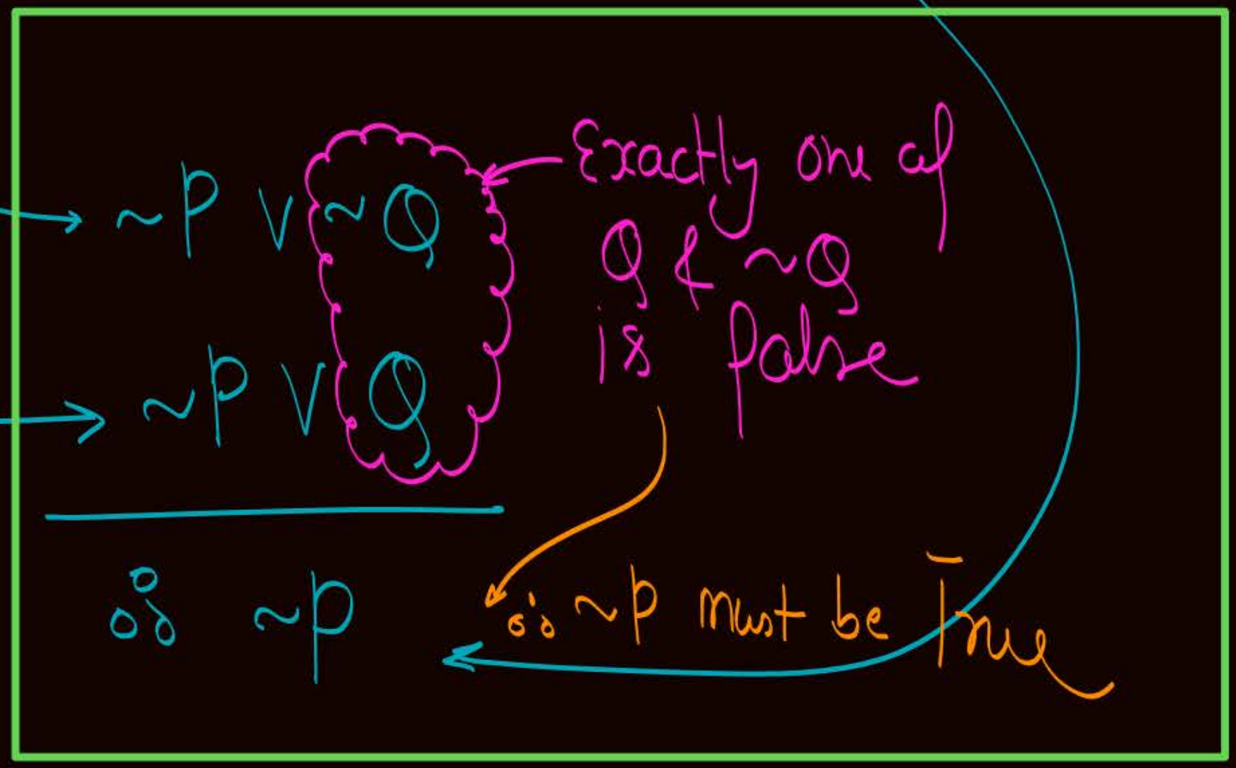
$\therefore \sim p$

is Valid { Modus Tollen }

* Q:. Check whether the following argument is valid or invalid

H.W.

$\{ p \rightarrow q ,\ \sim(p \wedge q) \}$ logically implies $\sim p$



Argument is Valid



Topic : Proof by contradiction

If we want to check whether the argument $\{P_1, P_2, P_3, \ldots P_n\} \vdash Q$ is valid or not

↳ then in order to check for the validity of the argument we will assume that the conclusion of the argument is false {i.e. we will assume that Q is false, i.e., we will assume that '$\sim Q$' is true}, and will include this false conclusion in the given set of premises.

∴ Our new set of Premises will be $\{P_1, P_2, P_3, \ldots, P_n, \sim Q\}$

- If this assumption leads to any Contradiction, then our assumption is wrong {we assumed that '$\sim Q$' is true and it is wrong means 'Q' is true}, and hence given argument is valid, if no contradiction, then invalid argument

Q:. Check whether the following argument is valid or invalid

$\{ p \rightarrow q , \sim(p \wedge q) \}$ logically implies $\sim p$

Solution using proof by Contradiction

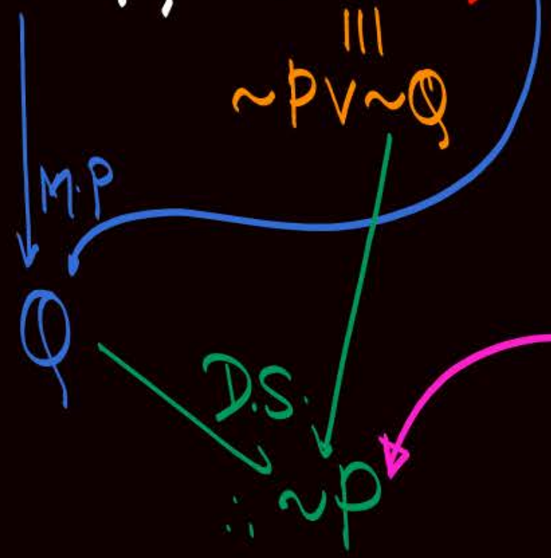
Conclusion of the argument is '$\sim p$'

① Assume that Conclusion of the argument is false
i.e, assume that '$\sim p$' is false, i.e, assume that 'p' is true

② Include this false Conclusion in the set of premises
∴ New set of premises will be

$\{ p \rightarrow q , \sim(p \wedge q) , p \}$

$\sim(p \wedge q) \equiv \sim p \vee \sim q$



it is Contradiction
∴ Our assumption is wrong
Hence Argument is Valid!

Q:. Check whether the following argument is valid or invalid

$\{ P \rightarrow Q, \sim(P \wedge Q) \}$ logically implies $\sim Q$

Solution using proof by Contradiction

① Assume Conclusion is false, i.e. '$\sim Q$' is false, i.e. 'Q' is true

② ∴ New set of premises will be

$\{ P \rightarrow Q, \sim(P \wedge Q), Q = T \}$

$\sim(P \wedge Q) \equiv \sim P \vee \sim Q$

$F \rightarrow T = T$

DS:

$\sim P = T \Rightarrow P = F$

at $P = F$, and $Q = T$ all three $P \rightarrow Q$, $\sim(P \wedge Q)$ & Q can be true together

∴ No Contradiction

& Hence Argument is invalid

Q:. Check whether the following argument is valid or not?

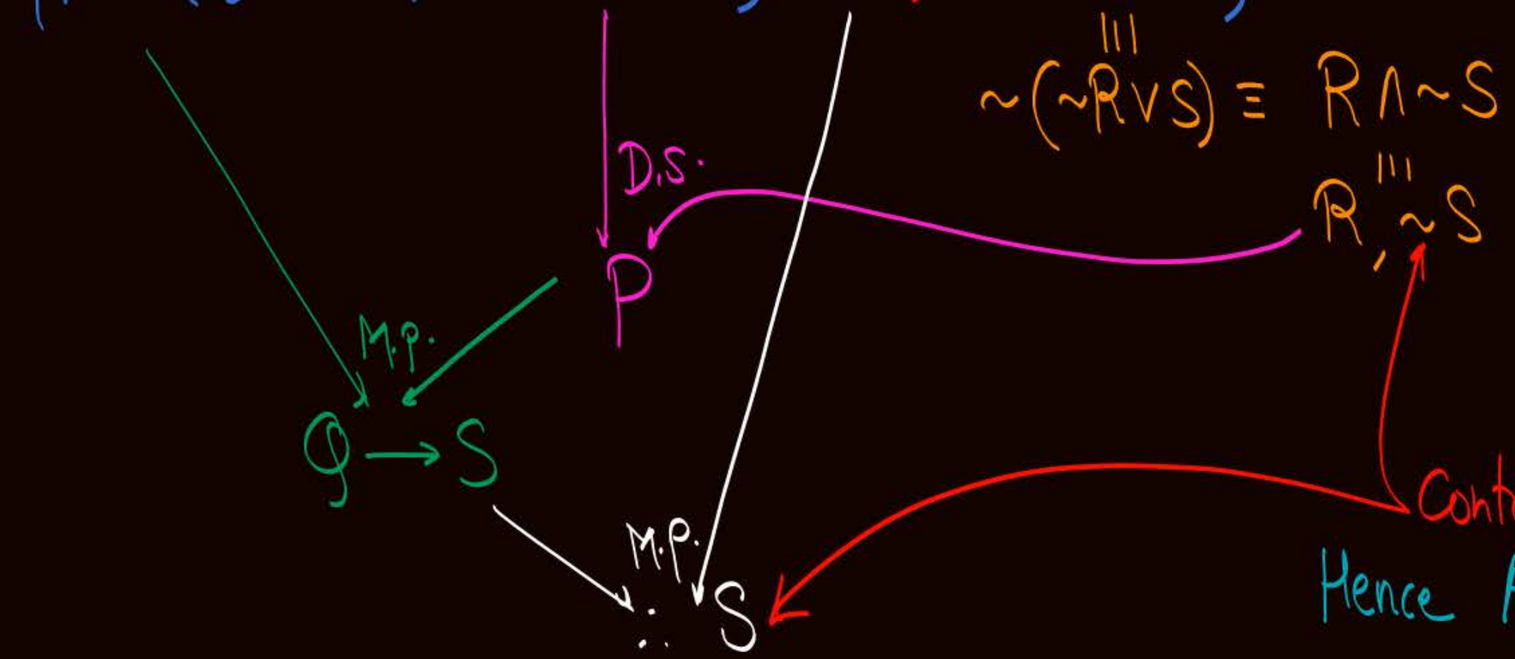
$\{P \to (Q \to S),\ \sim R \vee P,\ Q\}$ logically implies "$R \to S$"

① Assume Conclusion is False, i.e. assume $(R \to S)$ is False, i.e. assume $\sim(R \to S)$ is true

② ∴ Our new set of premises will be

$\{P \to (Q \to S),\ \sim R \vee P,\ Q,\ \sim(R \to S)\}$

$\sim(R \to S) \equiv \sim(\sim R \vee S) \equiv R \wedge \sim S$





Topic : Conditional proof

The argument

$\{P_1, P_2, P_3, \ldots P_n\} \vdash Q \rightarrow R$ is valid

if and only if, the argument

$\{P_1, P_2, P_3, \ldots P_n, Q\} \vdash R$ is valid

i.e, Checking whether the argument

$\{P_1, P_2, P_3, \ldots, P_n\} \vdash Q \rightarrow R$ is valid or not

is Exactly same as Checking

Whether the argument

$\{P_1, P_2, P_3, \ldots P_n, Q\} \vdash R$ is valid or not

If inclusion of Q into set of Premises results in any Contradiction then Argument is valid

Q:- Check whether the following argument is Valid or not?

Soln using Conditional proof

$\{P \rightarrow (Q \rightarrow S),\ \sim R \vee P,\ Q\}$ logically implies "$R \rightarrow S$"

$\{P \rightarrow (Q \rightarrow S),\ \sim R \vee P,\ Q\}$ logically implies "$R \rightarrow S$"

if and only if

$\{P \rightarrow (Q \rightarrow S),\ \sim R \vee P,\ Q,\ R\}$ logically implies S



Q:- Check whether the following argument is valid or invalid

If A works hard, then B or C will enjoy themselves $\equiv a \rightarrow (b \vee c)$

If B enjoys himself, then A will not work hard $\equiv b \rightarrow \sim a$

If C enjoy himself, then D will not enjoy himself $\equiv c \rightarrow \sim d$

$\therefore$ If A works hard then D will not enjoy himself $\equiv \therefore a \rightarrow \sim d$

Let, a : 'A' works hard

b : 'B' enjoy himself

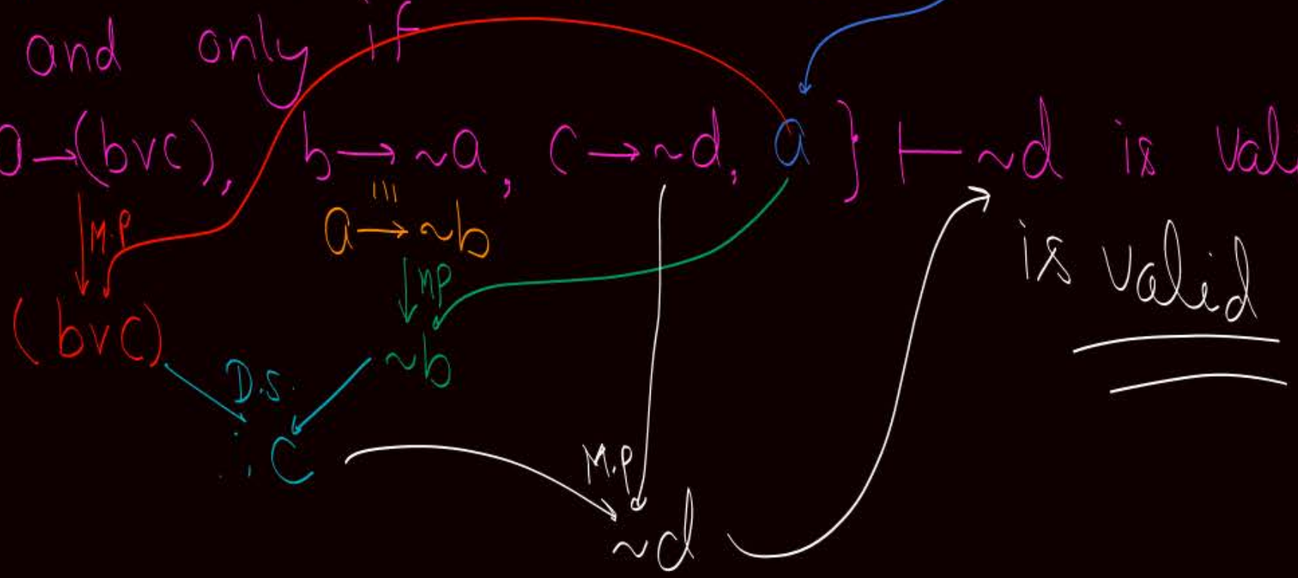
c : 'C' enjoy himself.

d : D enjoy himself.

$\{a \rightarrow (b \vee c),\ b \rightarrow \sim a,\ c \rightarrow \sim d\} \vdash a \rightarrow \sim d$ is valid

if and only if

$\{a \rightarrow (b \vee c),\ b \rightarrow \sim a,\ c \rightarrow \sim d,\ a\} \vdash \sim d$ is valid





Topic : Some important statements

1. $P \rightarrow Q$ $\equiv$ Q follows from P
2. P implies Q $\equiv P \rightarrow Q$
3. If P then Q $\equiv P \rightarrow Q$
4. P only if Q $\equiv P \rightarrow Q$
5. P is sufficient condition for Q $\equiv P \rightarrow Q$
6. Q is necessary condition for P $\equiv P \rightarrow Q$

eg: * $X \equiv$ you can crack GATE Exam only if you appear for the gate exam.

P : you can crack GATE Exam

Q : you appear for the gate exam.

$\sim Q$: you do not appear for gate

$\sim P$: you can not crack gate

$\equiv$ If you do not appear for gate then you can not crack gate

$\sim P \equiv \sim Q \rightarrow \sim P$

$\equiv P \rightarrow Q$

① $X \equiv$ If you crack the GATE exam, then you must have appeared for GATE exam

Correct

② $X \equiv$ If you appeared for GATE exam, then you crack GATE exam

Wrong



Topic : Some important statements

7. P if Q $\equiv q \rightarrow p$

8. P when Q $\equiv q \rightarrow p$

9. P follows from Q $\equiv q \rightarrow p$

10. P unless $\sim Q \equiv q \rightarrow p$

$\Downarrow\Downarrow$

$p \vee \sim q \equiv \sim q \vee p$

let p = you can crack gate
& q = you appear for gate

you can not crack Gate exam
Unless
you appear for gate exam

$\equiv$ $\sim p$ unless q $\equiv$ $\sim p \vee q$

$\Uparrow$

If you did not appear for gate
then you can not crack gate

$\equiv$ $\sim q \rightarrow \sim p$ $\Rightarrow$ $p \rightarrow q$

$$p \text{ unless } q \equiv p \vee q$$

Predicate Logic

(First order predicate logic)



Topic : Predicate

It is part of a sentence or clause stating something about the subject.

eg: Ram is a Sportsman

Ram — Subject

is a Sportsman — Predicate

] it is a proposition
it may be true
or
it may be false

* Let 'S' denotes "is a Sportsman"
Here 'S' is a predicate.

S(Ram) : Ram is a Sportsman

Predicate S will be applied over subject Ram

P: is a politician

S: is a sportsman

- P(x): x is a politician
- S(y): y is a sportsman

} x & y may be any person from the set of people under consideration

↑ it is our domain

$P(x) \rightarrow S(x)$: if P(x) then S(x)

: if x is a politician, then x is a sportsman

it may be true or it may be false

① if 'x' is not a politician, then implication will return true, does not matter whether x is a sportsman or not.

② If x is a politician then x must be sportsman as well for implication to return true.

③ If x is a politician but not a sportsman, then implication will return false

P: is a politician

S: is a sportsman

① $P(x) \rightarrow S(y)$: if x is a politician, then y is a sportsman

② $P(x) \vee S(y)$: x is a politician or y is a sportsman

③ $P(x) \wedge S(x)$: x is politician cum sportsman

④ $P(x) \leftrightarrow S(y)$: x is a politician if and only if y is a sportsman

if 'p' : is a Politician.

then '~p' : is not a Politician

P(Ram) ≡ Ram is a Politician.

~P(Ram) ≡ Ram is not a Politician.



Topic : Predicate with multiple subjects

Let 'F' : is a friend of

F(x, y) : x is a friend of y

- '~F' : is not a friend of

~F (Himanshu, Krishna) : Himanshu is not a friend of Krishna

Let G: is greater than

G(a, b): a is grater than b

{May be true or may be false}

eg: G(2, 3): is false

G(5, 5): is false

G(5, 3): True

eg: $F(x, y, t)$: Person 'x' can fool person 'y' at time 't'

$F(\text{Vishal}, \text{Himanshu}, 11{:}00\,\text{PM})$



2 mins Summary

- Topic: Practice questions
- Topic: Proof by contradiction
- Topic: Conditional proof
- Topic: Predicate logic

Slide

THANK - YOU